RF-Car

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Abstract

Keywords: CAN Bus, ISO 11898, Bus de campo, Arbitraje, STM32, Comunicación serial, CAN FD.

1 Introducción

2 Sistema Mínimo

2.1 STM32F103C8T6 - BluePill

2.2 TB6612FNG - H Bridge Driver

The TB6612FNG driver was used to control the vehicle's motors, as it allows for the management of two DC motors with speed and direction control. Here is shown the block diagram and pinout of the TB6612FNG:

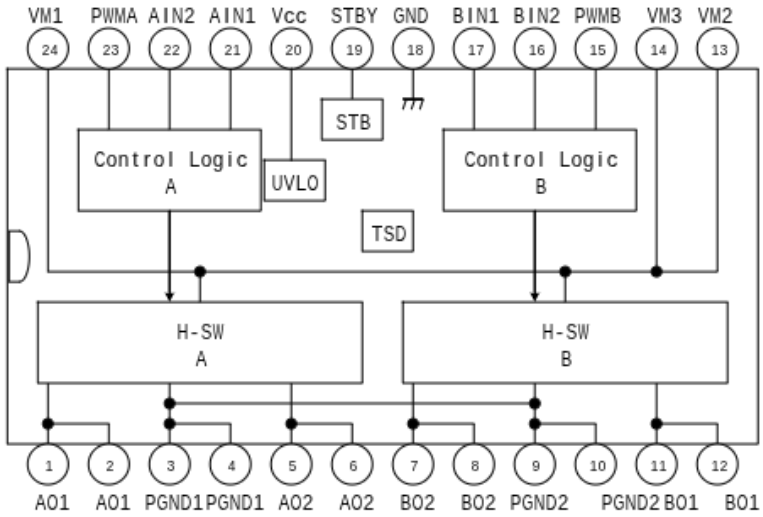


Fig. 1: Pinout del Driver TB6612FNG.

Additionaly, the functions of each pin of the TB6612FNG are integrated, as shown in Table 2.

Pin Functions

Pin NO.	Symbol	I/O	Remarks
1	AO1	O	chA output1
2	AO1		
3	PGND1	—	Power GND 1
4	PGND1		
5	AO2	O	chA output2
6	AO2		
7	BO2	O	chB output2
8	BO2		
9	PGND2	—	Power GND 2
10	PGND2		
11	BO1	O	chB output1
12	BO1		
13	VM2	—	Motor supply(2.5V ~ 13.5V)
14	VM3		
15	PWMB	I	chB PWM input / 200kΩ pull-down at internal
16	BIN2	I	chB input2 / 200kΩ pull-down at internal
17	BIN1	I	chB input1 / 200kΩ pull-down at internal
18	GND	—	Small signal GND
19	STBY	I	"L"=standby / 200kΩ pull-down at internal
20	Vcc	—	Small signal supply (2.7V ~ 5.5V)
21	AIN1	I	chA input1 / 200kΩ pull-down at internal
22	AIN2	I	chA input2 / 200kΩ pull-down at internal
23	PWMA	I	chA PWM input / 200kΩ pull-down at internal
24	VM1	—	Motor supply(2.5V ~ 13.5V)

Fig. 2: Tabla de funciones de cada Pin del Driver TB6612FNG.

The typical application circuit of the TB6612FNG is shown in Figure 3.

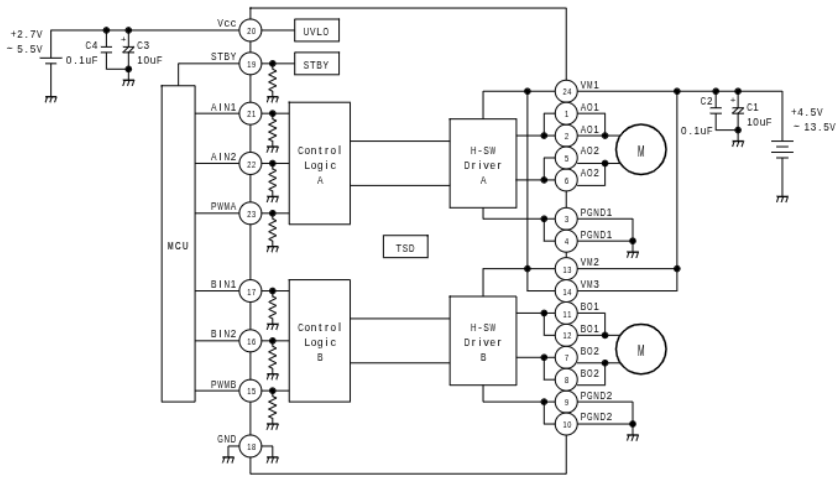


Fig. 3: Typical Application Circuit of the TB6612FNG.

We could see that the typical application circuit of the TB6612FNG is very simple, and it can be easily integrated into our project. By researching about the TB6612FNG Implementations, we found that the capacitor of 10 uF on VCC is not necessary, and it can be removed from the circuit. We could see that on the SparkFun breakout board, the capacitor of 10 uF is not present, as shown in Figure 4.

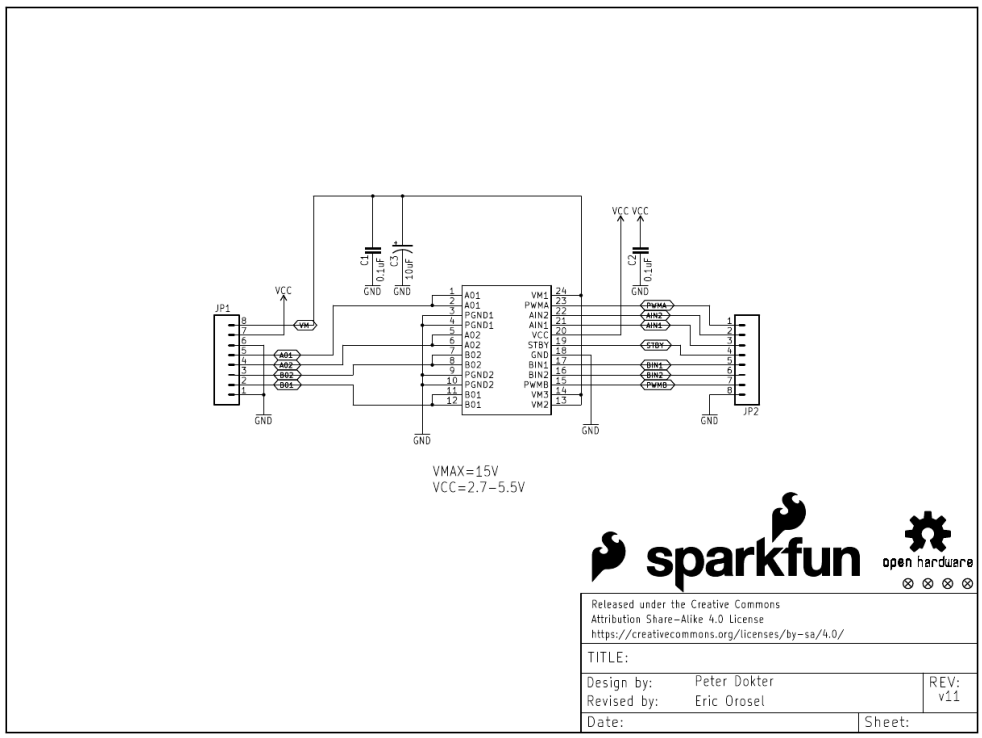


Fig. 4: SparkFun breakout board of the TB6612FNG.

The device uses two separate supplies:

- V_{CC}: logic supply, 2.7 V to 5.5 V.
- V_M: motor supply, up to 15 V.

Keeping both domains separate lets the microcontroller logic run from its own regulated rail while the motors are fed from a battery or power supply. Because the two rails have very different current profiles, each one needs its own decoupling, as discussed next.

2.3 Decoupling Capacitors

The typical application circuit in the datasheet places a 0.1 µF and a 10 µF capacitor on both V_M and V_{CC}. Each value has a different purpose.

Role of each rail.

The V_{CC} rail only feeds the control logic, so its current demand is a few milliamps and mostly constant. The V_M rail feeds the H-bridge output stage, which draws sharp current pulses at the PWM frequency and receives back-EMF spikes from the motor when the bridge switches or brakes. Capacitance on V_M is therefore the more critical of the two. Compact breakout boards often reduce the number of capacitors compared with the reference circuit, leaving extra bulk capacitance to the user.

Capacitor value and frequency.

A real capacitor behaves as a series RLC network 5: the capacitance C, the equivalent series resistance (ESR) and the equivalent series inductance (ESL).

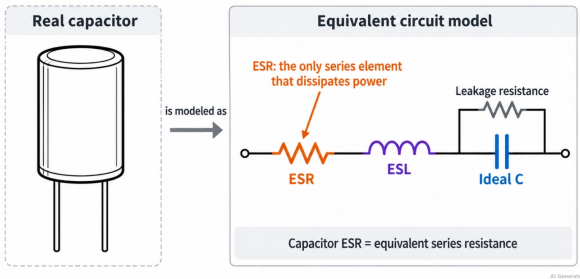


Fig. 5: Equivalent series resistance (ESR) and inductance (ESL) of a real capacitor.

Its impedance is

$$Z(f) = \text{ESR} + jw(\text{ESL}) + \frac{1}{jwC} = \text{ESR} + j\left(2\pi f\,\text{ESL} - \frac{1}{2\pi fC}\right),$$

$$Z(f) = \text{ESR} + j\left(2\pi f\,\text{ESL} - \frac{1}{2\pi fC}\right). \tag{1}$$

Given the self-resonant frequency f_{SRF} is where the inductive and capacitive **reactances** cancel each other, the capacitor behaves as a capacitor below f_{SRF} and as an inductor above it. So we could take the equation 1 to find the self-resonant frequency of a capacitor, which is given by:

$$\text{Im}(Z) = \text{Im}(\text{ESR} + j\left(2\pi f\,\text{ESL} - \frac{1}{2\pi fC}\right)) = 0 \Rightarrow 2\pi f\,\text{ESL} - \frac{1}{2\pi fC} = 0$$

Below the self-resonant frequency

$$f_{SRF} = \frac{1}{2\pi\sqrt{\mathrm{ESL} \cdot C}} \tag{2}$$

the capacitor acts as a capacitor; above it, the ESL dominates and it behaves as an inductor, which makes it useless for filtering. This is why two capacitors are used in parallel: the large one covers low frequencies and the small one covers high frequencies.

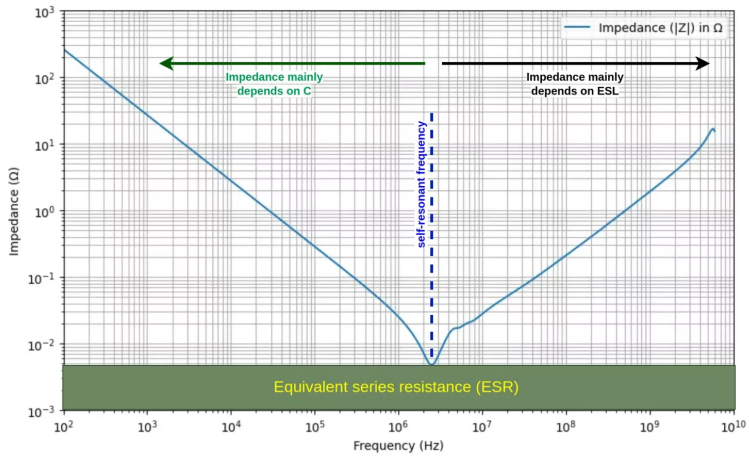


Fig. 6: Impedance of a real capacitor, showing the effect of ESR and ESL.

The ideal reactance $X_C = 1/(2\pi f C)$ at two frequencies of interest is:

Table 1: Ideal capacitive reactance of the two decoupling values.

C	X_C at 100 kHz	X_C at 10 MHz
10 µF	0.16 Ω	1.6 mΩ
0.1 µF	15.9 Ω	0.16 Ω

At the PWM frequency (up to 100 kHz) the 10 µF capacitor provides the low impedance path. At the tens of megahertz present in the switching edges, only the 0.1 µF capacitor remains effective, since its smaller ESL pushes its resonance much higher.

Capacitor size and voltage ripple.

The bulk capacitor supplies the current pulse of the bridge while the source cannot respond. A first-order estimate of the voltage droop on V_M is

$$\Delta V \approx \frac{I\,\Delta t}{C}, \tag{3}$$

where I is the pulse current and Δt its duration.

For example, with a 1A pulse lasting 5 us (half a period at 100 kHz) given capacitor values of 10 uF and 100 uF, the voltage droop is:

$$\Delta V = \frac{1\ \text{A} \cdot 5 \times 10^{-6}\ \text{s}}{10 \times 10^{-6}\ \text{F}} = 0.5\ \text{V} \quad \Delta V = \frac{1\ \text{A} \cdot 5 \times 10^{-6}\ \text{s}}{100 \times 10^{-6}\ \text{F}} = 0.05\ \text{V}$$

Ten times less with 100 µF. This is a worst-case estimate, since the supply also contributes current. Larger C reduces ripple, but increases the inrush current at power-up and the board area.

Package size (0402, 0603, 0805).

Surface-mount ceramic capacitors are named by their size code. In the imperial system, the codes correspond to the following dimensions (the metric code of the same part is different, e.g. imperial 0402 is metric 1005):

Table 2: Common imperial package sizes for ceramic capacitors.

Imperial code	Size (mm)	Typical ESL	Typical use here
0402	1.0×0.5	about 0.4 nH to 0.7 nH	0.1 µF, high-frequency bypass
0603	1.6×0.8	about 0.6 nH to 0.9 nH	0.1 µF to 1 µF, general purpose
0805	2.0×1.25	about 0.8 nH to 1.2 nH	1 µF to 10 µF, bulk



Fig. 7: SMD package sizes for ceramic capacitors.

The package affects the capacitor in three ways:

- **ESL.** A smaller body has a shorter current path, so its ESL is slightly lower and its self-resonant frequency slightly higher. In practice the difference is small, and the inductance of the PCB traces and vias to the capacitor is usually larger than the ESL of the part itself, so placement matters more than the package.
- **Capacitance and voltage available.** A given package can only hold a limited capacitance at a given voltage rating. A 0.1 µF capacitor is easy in 0402 or 0603, while 10 µF in ceramic generally needs 0805 or larger for a voltage rating suitable for V_M.
- **DC bias derating.** For the same value and voltage rating, a smaller package loses more of its capacitance when the DC voltage increases, and this is more severe in X5R/X7R dielectrics. A 10 µF 0603 part rated at 10 V can drop to a

fraction of its nominal value at 5 V bias, so the effective capacitance should be checked in the manufacturer's curves.

Larger packages are also more susceptible to cracking from board flex, which is a consideration for boards that are handled or mounted on vibrating platforms such as robots.

Ceramic versus electrolytic.

Table 3: Typical characteristics of the two capacitor types.

	Ceramic (MLCC)	Aluminum electrolytic
Typical value here	0.1 µF (up to 10 µF)	10 µF and above
ESR	Very low	Higher
ESL	Very low (about 1 nH)	Higher (order of 10 nH)
Self-resonance	Tens of MHz (for 0.1 µF)	Hundreds of kHz
Polarity	Non-polarized	Polarized
Main use	High-frequency noise	Bulk energy, low-frequency ripple
Caveat	Capacitance drops with DC bias (X5R/X7R)	Aging; voltage rating must be respected

With these typical values, a 0.1 µF ceramic with 1 nH of ESL resonates near 16 MHz, while a 10 µF electrolytic with 10 nH resonates near 500 kHz. The actual values depend on the package and manufacturer and should be taken from the capacitor datasheet.

Practical recommendations.

- Use 0402 or 0603 for the 0.1 µF ceramic and place it as close as possible to the V_{CC} and V_M pins, with short traces to ground.
- On V_M, add a bulk capacitor of 10 µF or more. If the motor supply is far from the board or the motors draw high current, use 47 µF to 100 µF. Equation 3 can be used to estimate the voltage droop.
- Choose a voltage rating well above the maximum V_M; for a 15 V maximum, a 25 V part is a common choice.

We conclude then the next values for the decoupling capacitors:

Table 4: Selected decoupling capacitors for the TB6612FNG.

Rail	Value	Type	Size	Voltage	Target frequency
V_{CC}	0.1 µF	Ceramic X7R	0603 (or 0402)	50 V	1 MHz to tens of MHz
V_M	0.1 µF	Ceramic X7R	0603 (or 0402)	50 V	1 MHz to tens of MHz
V_M	22 µF	Tantalum	1210	16 V	DC to about 1 MHz (PWM up to 100 kHz)

3 Conclusión

El protocolo CAN se ha consolidado durante casi cuatro décadas como el bus de comunicación de referencia en sistemas embebidos distribuidos, gracias a un diseño elegante que resuelve de forma determinista el acceso múltiple al medio mediante arbitraje bit a bit, y a un sistema de detección de errores excepcionalmente robusto para el bajo costo de su capa física. Si bien su ancho de banda y tamaño de carga útil resultan limitados frente a las demandas actuales de los vehículos autónomos y conectados, su evolución hacia CAN FD -y en desarrollo, CAN XL- demuestra la

vigencia del enfoque original. Para el trabajo práctico con microcontroladores STM32, el periférico bxCAN/FDCAN ofrece una vía directa y bien soportada por HAL para integrar nodos CAN en proyectos de instrumentación, robótica o redes de sensores distribuidos.